

Systems Analysis Study Guide

Lecture Notes - Visual Summary

RAD vs JAD (Lecture 2)

JAD Joint Application Development



Fact-Finding Technique

Team-based requirements gathering

- Brings users into development as **active participants**
- More expensive, can be cumbersome with large groups
- Results in **accurate requirements** statement
- Stronger commitment to system success

Focus: One phase (requirements modeling)

RAD Rapid Application Development



Complete Process

Prototyping & user involvement

- Users examine **working model early**
- Cuts development time & expense
- Limits cost of changes in long schedules
- Great for complex user interfaces

⚠ Weakness: May not emphasize strategic business needs



Key Difference: JAD focuses on **team-based fact-finding** (one phase), while RAD is a **compressed entire process**

Agile

Key Values



Individuals & Interactions

over ~~Processes & Tools~~



Working Software

over ~~Comprehensive Documentation~~



Customer Collaboration

over ~~Contract Negotiation~~



Responding to Change

over ~~Following a Plan~~

Agile = 4 Values + 12 Principles

Scrum Teams

Teams are **self-organized** and **self-directed**. Unlike other approaches, Scrum teams do not have a designated team leader.

Daily Stand-up Meeting (SCRUM)

1

What I did yesterday?

2

What I'll do today?

3

Any impediments I'm facing?

Lecture 3

Factors & Feasibility

External Factors



Technology



Government



Suppliers



Economy



Competitors

Internal Factors

Organizational structure, resources, culture, and strategic goals

Feasibility Study Types



Operational

Review of user needs, requirements, and expectations



Technical

Hardware, software, and network resources needed



Economic

Cost-benefit analysis and ROI



Schedule

Timeline feasibility assessment

Discretionary Projects

Management has a **choice** in implementing them

Non-discretionary Projects

No choice exists - must be done

Analysis Tools

PICK Chart: Find requests with low effort, high feasibility, and importance

Pareto Chart: Maps issues by descending frequency with cumulative curve



Project Manager

Responsible for overseeing all relevant tasks

Planning

Scheduling

Monitoring & Controlling

Reporting

Project Scheduling Tools



Gantt Chart

Horizontal bar chart with **tasks on vertical axis** and **time intervals on horizontal axis**



PERT/CPM

Bottom-up technique - analyzes complex projects as series of individual tasks

Work Breakdown Structure (WBS):

Breaking project into smaller tasks

Network Diagram:

Complete chain of task patterns for a project



Risk Management

Process of identifying, analyzing, anticipating, and monitoring risks

Brooks' Law: Adding more people to a project might actually **increase completion time**

System Analysis

⚠ It may be tempting to "just do something" but a systems project that **does not satisfy business requirements serves no useful purpose**

Requirements by Methodology



Structured Development → [Software Requirements Specification \(SRS\)](#)



Object Oriented → [Use Case Diagram](#)



Agile Development → [Personas & User Stories](#)

Systems Analyst Skills



Analytical Skills

Identify problems, evaluate key elements, develop useful solutions



Interpersonal Skills

Work with people at all organizational levels

Quick Reference

JAD

Joint Application Development - team-based fact-finding

RAD

Rapid Application Development - compressed full process

WBS

Work Breakdown Structure - breaking project into tasks

PERT

Program Evaluation Review Technique - complex project management

CPM

Critical Path Method - project scheduling

SRS

Software Requirements Specification - structured development doc